HRISHIN DEBNATH

Data Analyst | Student Researcher

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SUMMARY

Data analyst and student researcher: sole author of two air-quality papers under review and research intern at CreateBud. I work in Python, SQL, Power BI, Tableau, R and Excel. Looking for a data analyst internship.

SKILLS

Programming: Python (pandas, NumPy, SciPy, scikit-learn, statsmodels), R, SQL, DAX; working knowledge of JavaScript and TypeScript
ML & statistics: XGBoost, LightGBM, Random Forest, time-series forecasting, walk-forward validation, Optuna, hypothesis testing
Data & tools: PostgreSQL, MySQL, MongoDB, Power BI, Tableau, Excel, Power Query, Power Automate, Git, Jupyter

WORK EXPERIENCE

CreateBud Inc., Research Intern (remote, part-time) **Jul 2026 – Present**

- Tested CreateBud's pre-release co-working platform and reported 6+ bugs in its project, team and task features.

VibeWeave (early-stage startup), Market Analyst (remote, part-time) **Jun 2026 – Present**

- Built market studies of India's textile industry for five regions (fabric prices, raw-material commodities, listed companies) and a verified database of 26 clothing manufacturers.

Abelling, Data Annotator (remote, freelance) **Sep 2024 – Apr 2026**

- Annotated 5,000+ machine-learning data items on Encord, following detailed guidelines; earned performance bonuses.

Elevvo Pathways, Data Analyst Intern (remote, part-time) | **Work on GitHub** **Aug 2025 – Sep 2025**

- Segmented 1,998 retail customers with RFM analysis in Python; 306 "Champions" generated 23.81% of revenue.

Springer Capital, Data Automation Intern (remote, part-time) **Mar 2025 – Jun 2025**

- Maintained and fixed 10+ Microsoft Power Automate flows and ran sentiment analysis on 2,191 employee emails (TextBlob, VADER).

TechnoColabs Softwares, Data Analyst Intern (remote, part-time) **Sep 2024 – Nov 2024**

- Built Power BI loan-portfolio and borrower-risk dashboards on a company-provided dataset of 113,937 loans ($950M); 14.93% had defaulted or been charged off.

RESEARCH

Eid Holidays and Dhaka's Air: A Nine-Year Natural Experiment **Sep 2026**

Sole author. Columbia Junior Science Journal (under review) | **Code and data**

- Found PM2.5 falls 36% on average when Dhaka empties for Eid (95% CI 22% to 47%), across 16 holidays and 75,325 hourly U.S. Embassy readings, using an event study and 2,000 placebo runs.

Next-Day PM2.5 Forecasting in Dhaka with Gradient Boosting **Sep 2026**

Sole author. Submitted to Air Quality, Atmosphere & Health (under review) | **Code**

- Beat the persistence baseline by 5.22 µg/m³ RMSE (30.80 vs 36.02, R^2 0.795) on a 12-month holdout with XGBoost, using 248,953 hourly rows merged from OpenAQ, ERA5, NASA POWER and NASA FIRMS.

Bangla Medical Question Translation Study **Jul 2026 – Present**

Researcher, mentored study at CreateBud Inc. In preparation

- Building a Bangla version of the 12,000-question MedQA exam set; in a 500-question pilot of 5 open models, the revised prompt cut flagged translations for the chosen model from 399 to 100.

PROJECTS

Desk Watch: on-device study-focus app (in development) **Electron, TypeScript, MediaPipe**

- Windows app that uses the webcam, fully on the device, to sound an alarm when the user picks up a phone, leaves the desk or looks away; includes a site blocker, planner and Bangla interface.

SaaS Cohort Retention & Scenario Model | **GitHub** **Excel, Power Query, DAX**

- On a synthetic 17,585-row ledger, showed customer retention falls to about 70% by month 12 while revenue retention holds near 95%; 1,000 Monte Carlo runs gave 83% to 110% net revenue retention.

LEADERSHIP

CORViS (student-led non-profit), Deputy Director, Research & Development | **corvisbd.org** **Dec 2024 – Present**

- Organised 5+ seminars in the Speaks Academia series by inviting professors and industry professionals, including a panel on careers and AI in economics.

EDUCATION

St. Joseph Higher Secondary College, Dhaka: HSC, Science (English version) **2026 – 2028 (expected)**

WorldQuant University: Applied Data Science Lab | **Credential** **Dec 2025 – Jun 2026**

CERTIFICATIONS

Certificates: Google Data Analytics, **Google Business Intelligence**, **HackerRank SQL (Advanced)**, **freeCodeCamp Data Analysis with Python**, **Forage Quantium Data Analytics**